

AASTMT

Electrical Energy Department



Report in
Energy Environmental Aspects

***HEAT TRANSFER THROUGH
BUILDINGS***

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TABLE OF CONTENTS

Discussion.....	2
Best Case Building.....	4
Walls Calculations.....	5
Roof Calculations.....	6
Ground Calculations.....	7
Windows Calculations.....	7
Worst Case Building.....	8
Walls Calculations.....	9
Roof Calculations.....	10
Ground Floor Calculations.....	10
Windows Calculations.....	10
Most Sustainable Building.....	11
Walls Calculations.....	13
Roof Calculations.....	14
Ground Calculations.....	14
Windows Calculations.....	14

DISCUSSION

This report will be handling the heat transfer calculations through different materials and different building configurations and compare these values to try and find the best configuration for the most sustainable building. The report will be working on two buildings from the previous report: the one with the least solar radiation falling on it and the one with the highest solar radiation falling on it. The following data and tables will help in calculating the values we need for this study.

Solar Radiation Upon Vertical and Horizontal Surfaces in Summer for 30o N (watt/m2)										
Time/Orientation	Direct (Normal) Radiation									Diffused Radiation
	N	NE	E	SE	S	SW	W	NW	H	
07:00	155	530	600	315	0	0	0	0	345	35
08:00	90	515	635	385	0	0	0	0	560	40
09:00	35	405	555	380	0	0	0	0	745	45
10:00	0	255	410	325	45	0	0	0	900	50
11:00	0	90	220	220	90	0	0	0	1000	55
12:00	0	0	0	75	105	75	0	0	1025	55
13:00	0	0	0	0	90	220	220	90	1000	55
14:00	0	0	0	0	45	325	410	255	900	50
15:00	35	0	0	0	0	380	555	405	745	45
16:00	90	0	0	0	0	385	635	515	560	40
17:00	155	0	0	0	0	315	600	530	345	35

Figure 1. Solar Radiation Table for 30°N in Summer

This table has average values for solar radiation falling upon vertical and horizontal surfaces for an average day in summer for a location 30°N.

All the calculations done in this report will use data from the following tables as well. And the calculations will be done for two buildings from the previous study; the one with highest solar radiation, and the one with least solar radiation. The two buildings are shown in figures 4 and 5.

Thermal properties of some of the typical windows				Infiltration volume for doors	
Window Discription	U value (w/m2oC	Solar gain coefficient	Infiltration Volume		
			Wooden	Metal	
Single Glazing	4.75	0.8	2.25 (w), 3.65 (p), 0.5(H)	3.00 (w), 6.9 (p), 0.3 (H)	Wooden
Double Glazing	3.55	0.65			Metal
Double Glazing + Low-e & air filled	2.75	0.5			20.5 (P)
Double Glazing + Low-e & aragon filled	1.15	0.35			23.25 (P)
Triple Glazing + two Low-e coatings and Krypton filled	0.65	0.2			10.25 (W)
					14.9 (W)

Figure 2. Properties of windows

Internal surface resistance for building elements			
Surface	Resistance (m2oC/W)		
Walls	0.123		
Floors or ceilings	0.104		
Roofs	0.104		
External surface resistance for building elements			
Surface	Resistance (m2oC/W)		
	Sheltered	Normal	Severe
Walls	0.08	0.055	0.03
Floors or ceilings	0.07	0.045	0.02

- Sheltered includes the first three storeys of buildings in the interior of towns.
- Normal includes the fourth to eighth storeys of buildings in towns and most suburban and country premises.
- Severe includes ninth and higher floors of buildings in towns, fifth and higher floors in suburban districts and buildings exposed on coasts and hills.

Thermal properties of surfaces - Absorbance (a)		
Walls	White, glazed	0.25
	Light colours	0.4
	Dark colours	0.8
Roofs	Asphalt or bitumen	0.9
	Dark colours	0.8
	Red tiles	0.65
	White tiles	0.45
	Aluminium Oxidized	0.3
	Bright Aluminium	0.1

Figure 3. Properties of opaque surfaces

BEST CASE BUILDING

The best-case building is the one having the least solar radiation intensity falling on it according to the calculations done in the previous study, and it is shown in the figure below.

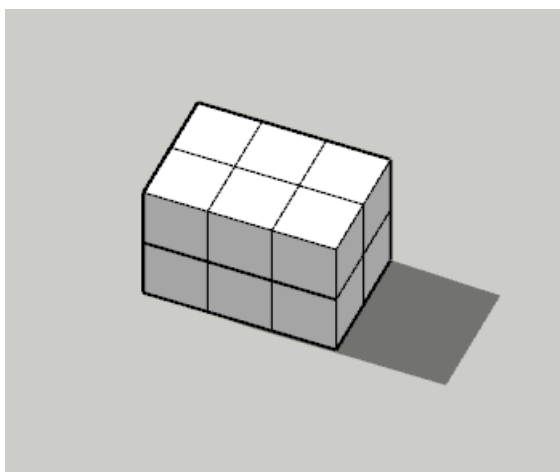


Figure 4. Best case building

To make this building the best case, the materials used for this building will be the best. So, it will have the best insulation and glazing for walls, roof, and windows. Since each cube of the cubes forming the building has a window in it, then the area when calculating the opaque surfaces must be the area of the opaque surface minus the area of the windows. The thermal resistance and transmittance values are calculated below.

$$\begin{aligned}
 R_{\text{wall}} &= 0.02 \times 1.88 + 0.12 \times 1.24 + 0.02 \times 1.88 \\
 &\quad 0.12 \times 1.24 + 0.12 \times 37 + 0.123 + 0.08 \\
 &= 5.0138 \text{ m}^2\text{C/W} \\
 U &= 0.199 \text{ W/m}^2\text{C} \\
 R_{\text{roof}} &= 0.07 + 0.104 + 0.02 \times 1.19 + 0.02 \times 1.88 \\
 &\quad + 0.05 \times 0.55 + 0.02 \times 37 + 0.02 \times 27 + 0.02 \times 0.69 \\
 &\quad + 0.02 \times 1.88 = 1.0943 \text{ m}^2\text{C/W} \\
 U &= 0.9138 \text{ W/m}^2\text{C}
 \end{aligned}$$

WALLS CALCULATIONS

For the highest insulated walls, the thermal transmittance (U-value) = $0.199 \text{ W/m}^2\text{C}$. The first-floor calculations will be as follows:

- Northern Façade:

$$Q_{op} = A \times U \times \Delta T$$

$$\Delta T = T_{sol} - T_{in}$$

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 80 \times 0.25 \times 0.08 = 36.6^\circ\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 3 - 1 \times 1 \times 3) \times 0.199 \times (36.6 - 25) = 55.4016 \text{ W}$$

- Eastern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.08 = 35.9^\circ\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 2) \times 0.199 \times (35.9 - 25) = 39.0438 \text{ W}$$

- Southern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.08 = 35.9^\circ\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 3 - 1 \times 1 \times 3) \times 0.199 \times (35.9 - 25) = 52.0584 \text{ W}$$

- Western Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 600 \times 0.25 \times 0.08 = 47^{\circ}\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 2) \times 0.199 \times (47 - 25) = 78.804 \text{ W}$$

And the ground-floor calculations will be as follows:

- Northern Façade:

$$Q_{op} = A \times U \times \Delta T$$

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 80 \times 0.25 \times 0.08 = 36.6^{\circ}\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 3 - 1 \times 1 \times 2) \times 0.199 \times (36.6 - 25) = 57.71 \text{ W}$$

- Eastern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.08 = 35.9^{\circ}\text{C}$$

$$Q_{op} = (3 \times 3 \times 2 - 1 \times 1) \times 0.199 \times (35.9 - 25) = 36.8747 \text{ W}$$

- Southern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.08 = 35.9^{\circ}\text{C}$$

$$Q_{op} = (3 \times 3 \times 3 - 1 \times 1 \times 2) \times 0.199 \times (35.9 - 25) = 54.2257 \text{ W}$$

- Western Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 600 \times 0.25 \times 0.08 = 47^{\circ}\text{C}$$

$$Q_{op} = (3 \times 3 \times 2 - 1 \times 1) \times 0.199 \times (47 - 25) = 74.429 \text{ W}$$

Therefore, the total heat transfer through all walls is the sum of all previous calculated values and is equal to:

$$Q_{op \text{ walls}} = 448.5472 \text{ W}$$

ROOF CALCULATIONS

For the highest insulated roof, the thermal transmittance (U-value) = 0.9138 W/m² °C.

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 790 \times 0.45 \times 0.07 = 59.885 \text{ }^{\circ}\text{C}$$

$$Q_{op \text{ roof}} = A \times U \times \Delta T = (6 \times 9) * 0.9138 * (59.885 - 25) = 1721.41 \text{ W}$$

GROUND CALCULATIONS

$$R_{ground} = 0.02 + \frac{1.19}{0.05} + 0.02 + 1.88 + 0.02 + 2 + 0.1 + 2.48 + 0.15 + 2.48 + 0.104 = 0.8529 \text{ m}^2\text{ }^{\circ}\text{C/W}$$

$$U_{ground} = 1.17 \text{ W/m}^2\text{ }^{\circ}\text{C}$$

$$Q_{op} = A \times U \times \Delta T$$

$$\Delta T = T_{soil} - T_{in}$$

$$T_{soil} = 20 \text{ }^{\circ}\text{C}$$

$$Q_{op} = A \times U \times \Delta T = (6 \times 9) * 1.17 * (20 - 25) = -315.9 \text{ W}$$

WINDOWS CALCULATIONS

- Heat gain/loss through conduction

$$Q_{cond} = A \times U \times \Delta T = (1 \times 1) * 0.65 * (35 - 25) = 6.5 \text{ W}$$

$$Q_{cond \text{ total}} = 6.5 * 12 = 78 \text{ W}$$

- Heat gain/loss through radiation

$$Q_{rad, \text{ north}} = A \times \theta \times I = (1 \times 1) * 0.2 * 80 = 16 \text{ W}$$

$$Q_{rad, east} = A \times \theta \times I = (1 \times 1) * 0.2 * 45 = 9 \text{ W}$$

$$Q_{rad, south} = A \times \theta \times I = (1 \times 1) * 0.2 * 45 = 9 \text{ W}$$

$$Q_{rad, west} = A \times \theta \times I = (1 \times 1) * 0.2 * 600 = 120 \text{ W}$$

- Heat gain/loss through infiltration

$$Q_{infil} = \text{Total crack length} \times \text{Infiltration volume} \times \Delta T \times 0.335$$

$$= (2 \times 1 + 2 \times 1) * 0.3 * (35 - 25) * 0.335 = 4.02 \text{ W}$$

Therefore, the total heat transfer through all windows is the sum of all 3 forms of heat transfer through each window which is equal to:

$$Q_{windows} = 78 + (5 * 16) + (1 * 9) + (5 * 9) + (1 * 120) + (4.02 * 12) = 330.24 \text{ W}$$

After concluding all these values, the total heat transfer through all surfaces of this building can be as follows:

$$Q_{Total} = Q_{walls} + Q_{roof} + Q_{ground} + Q_{windows} = 448.5472 + 1721.41 - 315.9 + 330.24 = 2184.2972 \text{ W}$$

Hence the power of the cooling unit that can be used for this building is

$$P = \frac{2184.2972}{746} = 2.9801 \approx 3 \text{ HP}$$

WORST CASE BUILDING

The worst-case building is the one having the highest solar radiation intensity falling on it according to the calculations done in the previous study, and it is shown in the figure below.

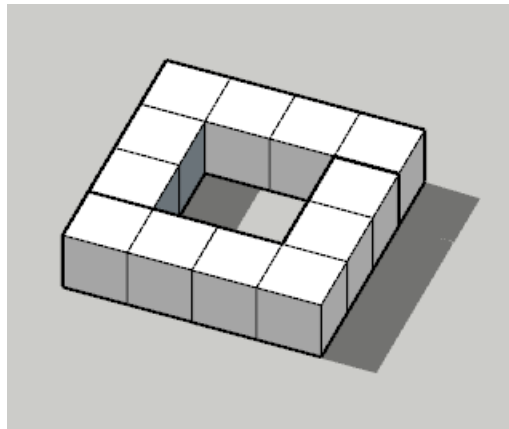


Figure 5. Worst case building

In order to make this building the worst case, the materials used for this building will be the worst. So, it will have the worst insulation and glazing for walls, roof, and windows. Since each cube of the cubes forming the building has a window in it, then the area when calculating the opaque surfaces must be the area of the opaque surface minus the area of the windows. The values of thermal resistance and transmittance are calculated below.

$$\begin{aligned}
 R_{\text{wall}} &= 0.123 + 0.08 + 0.02 \times 1.88 + 0.12 \times 0.68 + \\
 &\quad 0.02 \times 1.88 = 0.3598 \\
 U_{\text{wall}} &= 2.779 \text{ W/m}^2\text{.}^\circ\text{C} \\
 R_{\text{roof}} &= 0.02 \times 2 + 0.07 + 0.104 + 0.02 \times 1.19 + \\
 &\quad 0.02 \times 1.88 + 0.05 \times 0.55 + 0.02 \times 0.69 \\
 &\quad + 0.02 \times 1.88 = 0.3543 \text{ m}^2\text{.}^\circ\text{C/W} \\
 U_{\text{roof}} &= \frac{1}{R} = 2.82246 \text{ W/m}^2\text{.}^\circ\text{C}
 \end{aligned}$$

WALLS CALCULATIONS

For walls with lowest insulation value, the thermal transmittance (U-value) = 2.779W/m²°C. The calculations will be as follows:

- Northern Façade:

$$Q_{op} = A \times U \times \Delta T$$

$$\Delta T = T_{sol} - T_{in}$$

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 80 \times 0.8 \times 0.08 = 40.12^\circ\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 6 - 1 \times 1 \times 6) \times 2.779 \times (40.12 - 25) = 2016.887 \text{ W}$$

- Eastern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.8 \times 0.08 = 37.88^\circ\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 6 - 1 \times 1 \times 4) \times 2.779 \times (37.88 - 25) = 1932.85 \text{ W}$$

- Southern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.8 \times 0.08 = 37.88^\circ\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 6 - 1 \times 1 \times 2) * 2.779 * (37.88 - 25) = 1861.263 \text{ W}$$

- Western Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 600 \times 0.8 \times 0.08 = 73.4 \text{ }^{\circ}\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 6) * 2.779 * (73.4 - 25) = 7263.19 \text{ W}$$

Therefore, the total heat transfer through all walls is the sum of all previous calculated values and is equal to:

$$Q_{op \text{ walls}} = 13074.194 \text{ W}$$

ROOF CALCULATIONS

For the roof with the least insulation value, the thermal transmittance (U-value) = 2.822 W/m²°C.

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 790 \times 0.8 \times 0.07 = 79.24 \text{ }^{\circ}\text{C}$$

$$Q_{op \text{ roof}} = A \times U \times \Delta T = 108 * 2.822 * (79.24 - 25) = 16531.05 \text{ W}$$

GROUND FLOOR CALCULATIONS

$$R_{ground} = 0.02 * 1.19 + 0.02 * 1.88 + 0.05 * 0.55 + 0.02 * 2 + 0.1 * 2.48 + 0.15 * 3.48 + 0.104 = 0.8529 \text{ m}^2\text{ }^{\circ}\text{C/W}$$

$$U_{ground} = 1.17 \text{ W/m}^2\text{ }^{\circ}\text{C}$$

$$Q_{op} = A \times U \times \Delta T$$

$$\Delta T = T_{soil} - T_{in}$$

$$T_{soil} = 20 \text{ }^{\circ}\text{C}$$

$$Q_{ground} = A \times U \times \Delta T = 108 * 1.17 * (20 - 25) = -631.8 \text{ W}$$

WINDOWS CALCULATIONS

- Heat gain/loss through conduction

$$Q_{cond} = A \times U \times \Delta T = (1 \times 1) * 4.75 * (35 - 25) = 47.5 \text{ W}$$

$$Q_{cond \text{ total}} = 47.5 * 12 = 570 \text{ W}$$

- Heat gain/loss through radiation

$$Q_{rad, \text{ north}} = A \times \theta \times I = (1 \times 1) * 0.8 * 80 = 64 \text{ W}$$

$$Q_{rad, \text{ east}} = A \times \theta \times I = (1 \times 1) * 0.8 * 45 = 36 \text{ W}$$

$$Q_{rad, \text{ south}} = A \times \theta \times I = (1 \times 1) * 0.8 * 45 = 36 \text{ W}$$

- Heat gain/loss through infiltration

$$Q_{infil} = \text{Total crack length} \times \text{Infiltration volume} \times \Delta T \times 0.335$$

$$= (2 \times 1 + 2 \times 1) * 6.9 * (35 - 25) * 0.335 = 92.46 \text{ W}$$

Therefore, the total heat transfer through all windows is the sum of all 3 forms of heat transfer through each window which is equal to:

$$Q_{windows} = 570 + (6 * 64) + (4 * 36) + (2 * 36) + (92.46 * 12) = 2279.52 \text{ W}$$

After concluding all these values, the total heat transfer through all surfaces of this building can be as follows:

$$Q_{Total} = Q_{walls} + Q_{roof} + Q_{ground} + Q_{windows} = 13074.194 + 16531.05 + 2279.52 - 631.8 = 31252.964 \text{ W}$$

Hence the power of the cooling unit that can be used for this building is

$$P = \frac{31252.964}{746} = 41.89 \approx 42 \text{ HP}$$

MOST SUSTAINABLE BUILDING

The most sustainable building was designed to have the least possible solar radiation intensity falling on its facades. The design of the building depends on the idea of self-shading, where most of the roof area has shaded as well as the western façade and those two facades have the highest intensity falling on them at 3:00pm. The total solar radiation falling on the envelop of this building was calculated and found to be 53910 W. The building configuration is shown in the figure below.

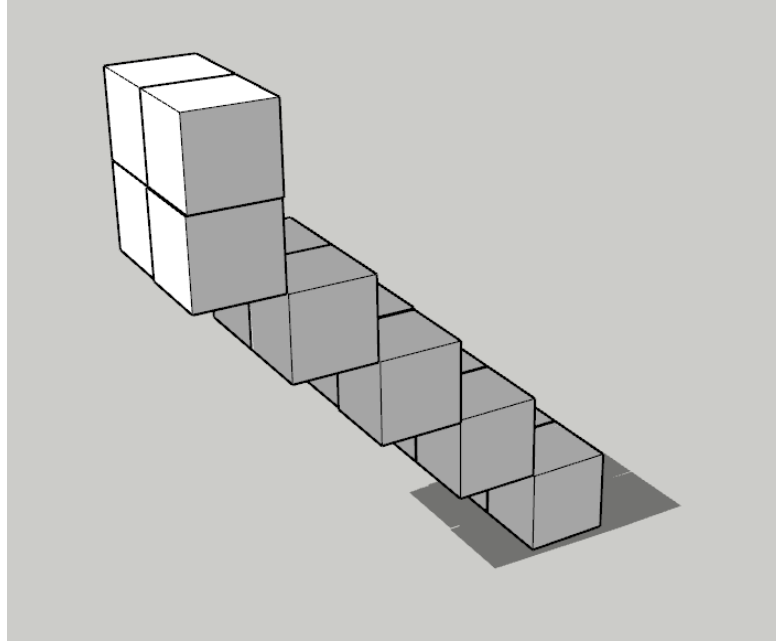


Figure 6. Most sustainable building

To make this building the most sustainable, the materials used for this building will be the best. So, it will have the best insulation which is polyurethane as the thermal insulation (PUR) having a resistivity of $R = 43.5 \text{ m}^2\text{°C/W}$ and Bitumen as the water insulation. White glazing was chosen for walls, and windows are triple glazed. The calculations for the overall thermal resistance and transmittance (U-value) for walls, roofs and ground are shown below.

➤ First three storeys

$$R_{\text{wall}} = 0.123 + 0.08 + 0.02 \times 1.88 + 0.12 \times 1.24 \\ + 0.22 \times 43.5 + 0.12 \times 1.24 + 0.02 \times 1.88 \\ = 10.1458 \text{ m}^2\text{°C/W}$$

$$U_{\text{wall}} = 0.09856 \text{ W/m}^2\text{°C}$$

➤ Other three storeys

$$R_{\text{wall}} = 0.055 + 0.123 + 0.02 \times 1.88 + 0.12 \times 1.24 + 0.22 \times 43.5 + 0.12 \times 1.24 + 0.02 \times 1.88 = 10.1208 \text{ m}^2\text{°C/W}$$

$$U_{\text{wall}} = 0.0988 \text{ W/m}^2\text{°C}$$

$$R_{\text{Roof}} = 0.045 + 0.104 + 0.02 \times 1.19 + 0.02 \times 1.88 + 0.05 \times 0.55 + 0.02 \times 43.5 + 0.02 \times 1.88 + 0.02 \times 0.69 + 0.02 \times 2 = 1.1993 \text{ m}^2\text{°C/W}$$

$$U_{\text{Roof}} = 0.8338 \text{ W/m}^2\text{°C}$$

➤ Ground

$$R_{\text{Ground}} = 0.02 \times 1.19 + 0.02 \times 1.88 + 0.05 \times 0.55 + 0.1 \times 2.48 + 0.02 \times 2 + 0.15 \times 2.48 + 0.104 = 1.2104 \text{ m}^2\text{°C/W}$$

$$U_{\text{Ground}} = 0.82617 \text{ W/m}^2\text{°C}$$

Since each cube of the cubes forming the building has a window in it, then the area when calculating the opaque surfaces must be the area of the opaque surface minus the area of the windows.

WALLS CALCULATIONS

The first 3 storeys, the calculations will be as follows:

- Northern Façade:

$$Q_{op} = A \times U \times \Delta T$$

$$\Delta T = T_{\text{sol}} - T_{\text{in}}$$

$$T_{\text{sol}} = T_{\text{out}} + I \times a \times R_{\text{out}} = 35 + 80 \times 0.25 \times 0.08 = 36.6 \text{ °C}$$

$$\therefore Q_{op} = (3 \times 3 \times 3) \times 0.09856 \times (36.6 - 25) = 30.86 \text{ W}$$

- Eastern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.08 = 35.9^{\circ}\text{C}$$

$$\therefore Q_{op} = (3 \times 6 \times 3 - 1 \times 1 \times 6) \times 0.09856 \times (35.9 - 25) = 51.56 \text{ W}$$

- Southern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.08 = 35.9^{\circ}\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 3) \times 0.09856 \times (35.9 - 25) = 29 \text{ W}$$

- Western Façade:

$$T_{sol, diff} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.08 = 35.9^{\circ}\text{C}$$

$$\therefore Q_{op} = (3 \times 6 \times 3) \times 0.09856 \times (35.9 - 25) = 58.01 \text{ W}$$

And the other three storeys, calculations will be as follows:

- Northern Façade:

$$Q_{op} = A \times U \times \Delta T$$

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 80 \times 0.25 \times 0.055 = 36.1^{\circ}\text{C}$$

$$\therefore Q_{op} = (3 \times 3 \times 3) \times 0.0988 \times (36.1 - 25) = 29.61 \text{ W}$$

- Eastern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.055 = 35.6^{\circ}\text{C}$$

$$Q_{op} = (3 \times 6 \times 3 - 1 \times 1 \times 6) \times 0.0988 \times (35.6 - 25) = 50.26 \text{ W}$$

- Southern Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.055 = 35.6^{\circ}\text{C}$$

$$Q_{op} = (3 \times 3 \times 3) \times 0.0988 \times (35.6 - 25) = 28.27 \text{ W}$$

- Western Façade:

$$T_{sol} = T_{out} + I \times a \times R_{out} = 35 + 600 \times 0.25 \times 0.055 = 43.25^{\circ}\text{C}$$

$$\therefore Q_{op} = (6 \times 6) * 0.0988 * (43.25 - 25) = 64.9 \text{ W}$$

$$T_{sol, diff} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.25 \times 0.055 = 35.6^\circ\text{C}$$

$$\therefore Q_{op} = (3 \times 6) * 0.0988 * (35.6 - 25) = 18.85 \text{ W}$$

Therefore, the total heat transfer through all walls is the sum of all previous calculated values and is equal to:

$$Q_{op \text{ walls}} = 361.32 \text{ W}$$

ROOF CALCULATIONS

The first 3 storeys calculations will be as follows:

$$T_{sol, diff} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.45 \times 0.07 = 36.4^\circ\text{C}$$

$$Q_{op \text{ roof}} = A \times U \times \Delta T = (6 \times 3 \times 3) * 0.8168 * (36.4 - 25) = 502.82 \text{ W}$$

The other three storeys calculations will be as follows:

$$T_{sol, diff} = T_{out} + I \times a \times R_{out} = 35 + 45 \times 0.45 \times 0.045 = 35.9^\circ\text{C}$$

$$Q_{op \text{ roof}} = A \times U \times \Delta T = (6 \times 3) * 0.8338 * (35.9 - 25) = 163.59 \text{ W}$$

$$T_{sol, diff} = T_{out} + I \times a \times R_{out} = 35 + 600 \times 0.45 \times 0.045 = 47.15^\circ\text{C}$$

$$Q_{op \text{ roof}} = A \times U \times \Delta T = (6 \times 3) * 0.8338 * (47.15 - 25) = 332.43 \text{ W}$$

$$\therefore Q_{roof \text{ total}} = 998.84 \text{ W}$$

GROUND CALCULATIONS

$$Q_{op} = A \times U \times \Delta T$$

$$\Delta T = T_{soil} - T_{in}$$

$$T_{soil} = 20^\circ\text{C}$$

$$Q_{op} = A \times U \times \Delta T = (6 \times 3) * 0.82617 * (20 - 25) = -74.36 \text{ W}$$

WINDOWS CALCULATIONS

- Heat gain/loss through conduction

$$Q_{cond} = A \times U \times \Delta T = (1 \times 1) * 0.65 * (35 - 25) = 6.5 \text{ W}$$

$$Q_{cond \text{ total}} = 6.5 * 12 = 78 \text{ W}$$

- Heat gain/loss through radiation

$$Q_{rad, \text{ east}} = A \times \theta \times I = (1 \times 1) * 0.2 * 45 = 9 \text{ W}$$

- Heat gain/loss through infiltration

$$Q_{infil} = \text{Total crack length} \times \text{Infiltration volume} \times \Delta T \times 0.335$$

$$= (2 \times 1 + 2 \times 1) * 0.3 * (35 - 25) * 0.335 = 4.02 \text{ W}$$

Therefore, the total heat transfer through all windows is the sum of all 3 forms of heat transfer through each window which is equal to:

$$Q_{windows} = 78 + (9 * 12) + (4.02 * 12) = 234.24 \text{ W}$$

After concluding all these values, the total heat transfer through all surfaces of this building can be as follows:

$$Q_{Total} = Q_{walls} + Q_{roof} + Q_{ground} + Q_{windows} = 361.32 + 998.84 - 74.36 + 234.24 = 1520.04 \text{ W}$$

Hence the power of the cooling unit that can be used for this building is

$$P = \frac{1520.04}{746} = 2 \text{ HP}$$